Crop Rotations under Drought in the Valley of Mexico

Satellite data, a crop model with water and nitrogen, and the Mexican rainfed context

Gradient Descent Farmers — NASA Space Apps warm-up
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Abstract

We want to compare crop sequences (rotations) for rainfed farms in central Mexico, where the rainy season is short, variable and interrupted by a mid-summer dry spell (the *canícula*). We built a pipeline that pulls free NASA data (POWER weather, SMAP soil moisture, GLDAS and IGBP-DIS soil properties) for any point, and drives the WOFOST 8.1 crop model with it, adding nitrogen, frost, rain-onset sowing, fertilizer practice from a Texcoco field trial, and season-to-season carryover of soil water and nitrogen. At CIMMYT's El Batán station (Texcoco) over 2005–2025, the simulated fertilized maize yields (≈6–7 t/ha dry matter) are in the range of field trials at Texcoco (4.8–6.5 t/ha). Nitrogen, not water, is the largest yield gap in most years; rotating maize with faba bean raises maize yields mostly in dry years; and frost and late sowing shorten the season more than expected. This document records the data, equations, assumptions and results so the work can be reproduced and challenged.

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1 Goal and approach

Decision problem. Which crop to grow each rainy season, and in which order over the years, on a rainfed (*temporal*) farm in the Mexican highlands. The answer depends on the weather of each year, which nobody knows when sowing, so we evaluate rotations over many real past years and under drier scenarios.

Approach.

1. **Data** (Section 3): for a point (latitude, longitude), pull daily weather, soil moisture and static soil properties from NASA sources, and check that they are read correctly and behave physically.
2. **Crop model** (Section 4): simulate each crop day by day with WOFOST 8.1 through the Python package PCSE, at three production levels: potential, water-limited and water- and nitrogen-limited.
3. **Mexican context** (Section 5): sowing when the rains start, local fertilizer practice, legume nitrogen fixation, frost, drought indicators and drier-climate scenarios.
4. **Rotations** (Section 6): chain seasons so water and nitrogen left by one crop affect the next, and compare rotations over 2005–2025.

2 Study site

The default site is CIMMYT's El Batán research station in Texcoco, Estado de México (19.53 °N, 98.85 °W, ≈2,250 m), in the farmland northeast of Mexico City, where long-term rainfed maize trials are run. The climate has a rainy season from May to October and a dry, cool season from November to April with occasional frost. In NASA POWER data, May–October rain averaged 774 mm over 2005–2025 (minimum 478 mm in 2023, maximum 1,005 mm in 2025), and June rain averaged 145 mm (12 mm in 2023). The soil at the site is a loam (GLDAS), with field capacity 0.373 m^3/m^3, wilting point 0.137 m^3/m^3 (IGBP-DIS) and porosity 0.437 (GLDAS), i.e. 236 mm of plant-available water in the top metre.

3 Data

3.1 Reading SMAP efficiently

Each SMAP granule is a ≈140 MB HDF5 file covering the globe for one 3-hour window. Every variable is stored as one gzip-compressed chunk of 1624×3856 cells (≈4.3 MB compressed), so even reading one pixel downloads the whole chunk. Instead, `smap.py` asks NASA's OPeNDAP server

Table 1: Data sources. All are free; SMAP and IGBP-DIS need a NASA Earthdata login.

Source	What it provides	Resolution	Format	Script
NASA POWER (NASA Langley Research Center, 2026)	Daily weather: min/max temperature, rain (bias-corrected), solar radiation, humidity, wind	0.5°×0.625°	JSON (API)	`power.py`
SMAP L4 SPL4SMGP v8 (Reichle et al., 2025)	3-hourly soil moisture (0–5 cm, 0–100 cm, profile), temperature, fluxes; 45 variables	9 km EASE-Grid 2.0	HDF5	`smap.py`
GLDAS soils (Rodell et al., 2004)	Texture class, sand/silt/clay, porosity	0.25°	netCDF4	`gldas_soils.py`
IGBP-DIS (Global Soil Data Task, 2000)	Field capacity, wilting point, available water, bulk density, soil C and N (0–100 cm)	1/12° (≈9 km)	ASCII grid	`igbp_soils.py`

for only the pixel (`/Geophysical_Data/<var>[row][col]`), 16 requests in parallel, falling back to streaming the HDF5 if OPeNDAP fails. This cut one week of data from ≈25 min to ≈30 s. The pixel is the cell whose centre is nearest to the point; because EASE-Grid 2.0 is cylindrical, latitude is constant along rows and longitude along columns, so the row and column are found from two 1-D arrays:

$$i^* = \arg\min_i |\varphi_i - \varphi|, \qquad j^* = \arg\min_j |\lambda_j - \lambda|. \tag{1}$$

3.2 Validation of the data pipeline

At Celaya, Guanajuato, for all of 2024 (2,928 three-hourly steps) we checked:

- OPeNDAP values equal direct HDF5 reads to 6 significant digits;
- the 1-D pixel lookup equals a full 2-D nearest search at 5 test sites (cell centres 2.8–4.8 km away);
- no missing or duplicate time steps; soil moisture within physical bounds;
- SMAP's precipitation input and POWER rain agree on the annual total (728 vs 727 mm), correlate daily ($r = 0.69$) and agree on rain/no-rain days 94% of the time; surface soil moisture rises 0.019 m^3/m^3 on days with > 5 mm of POWER rain and falls on dry days;
- monthly SMAP land-surface temperature and POWER air temperature correlate at $r = 0.99$.

SMAP L4 meets its accuracy requirement of 0.04 m^3/m^3 unbiased RMSE against in-situ sensors (Reichle et al., 2017). Root-zone percentiles (`sm_rootzone_pctl`) are missing in NASA's files for about a third of 2024 at Celaya, in dry-season blocks. SMAP data from 14 May to 28 July 2026 are being reprocessed after a geolocation error (National Snow and Ice Data Center, 2026).

4 Crop model

We use WOFOST 8.1 (de Wit et al., 2019; Supit et al., 1994) through PCSE 6.0.13 (de Wit, 2024a), with crop parameters from the WOFOST crop parameter repository (de Wit, 2024b). WOFOST simulates one crop on one field, one day at a time. Three model configurations give the three production levels used for yield-gap analysis:

Level	PCSE model	Limiting factors
Potential Y_p	`Wofost81_PP`	temperature, radiation, CO_2
Water-limited Y_w	`Wofost81_WLP_CWB`	+ soil water (classic water balance)
Water- and N-limited Y_{wn}	`Wofost81_NWLP_CWB_CNB`	+ nitrogen (classic N balance)

Below, t is the day, T the mean daily temperature, and names in `TYPEWRITER` are WOFOST parameters. A parameter ending in `TB` is a table $\{(x_k, y_k)\}$ evaluated by linear interpolation.

4.1 Weather: reference evapotranspiration

PCSE computes the FAO-56 Penman–Monteith reference evapotranspiration (Allen et al., 1998) from POWER data:

$$ET_0 = \frac{\Delta\, R_n + \gamma\, \dfrac{900}{T+273}\, u_2\, (e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\, u_2)}, \tag{2}$$

with R_n the net radiation (as mm of water), Δ the slope of the saturation vapour pressure curve, γ the psychrometric constant (which depends on air pressure, hence elevation), u_2 wind at 2 m, and $e_s - e_a$ the vapour pressure deficit.

4.2 Phenology

The development stage D runs from 0 (emergence) through 1 (flowering) to 2 (maturity) and depends on temperature only:

$$\frac{dD}{dt} = \begin{cases} \texttt{DTSMTB}(T)/\texttt{TSUM1}, & D < 1, \\ \texttt{DTSMTB}(T)/\texttt{TSUM2}, & 1 \le D < 2. \end{cases} \tag{3}$$

For the maize variety used, `DTSMTB` rises linearly from 0 at 6 °C to 24 at 30 °C, `TSUM1`=695 and `TSUM2`=860 °C d.

4.3 Photosynthesis

Leaf photosynthesis saturates with absorbed light I:

$$A(I) = A_{\max}\left(1 - e^{-\varepsilon I / A_{\max}}\right), \tag{4}$$

with $\varepsilon = \texttt{EFF}$. In WOFOST 8.1 the maximum rate depends on leaf nitrogen. The specific leaf N at cumulative leaf area L from the top of the canopy is

$$\mathrm{SLN}(L) = N_{LV}\, \frac{k_N\, e^{-k_N L}}{1 - e^{-k_N\, \mathrm{LAI}}}, \qquad A_{\max}(L) = f_{CO_2}\, f_T\, \min\!\big(A_{\mathrm{ref}},\ \max(0,\ s\,(\mathrm{SLN}(L) - N_b))\big), \tag{5}$$

with N_{LV} the leaf N (kg/ha), $k_N = \texttt{KN}$, $A_{\mathrm{ref}} = \texttt{AMAX_REF}$, $s = \texttt{AMAX_SLP}$, $N_b = \texttt{AMAX_LNB}$. Light extinction through the canopy follows e^{-kL} for diffuse and direct light, with sunlit and shaded leaves treated separately (Spitters et al., 1986). Canopy photosynthesis is integrated with 3-point Gaussian quadrature over canopy depth and over the day:

$$G = \mathrm{DAYL} \sum_{i=1}^{3} w_i\, \mathrm{LAI} \sum_{j=1}^{3} w_j\, A_{ij}, \qquad x = (0.113,\ 0.5,\ 0.887),\ w = (0.278,\ 0.444,\ 0.278), \tag{6}$$

where A_{ij} is the leaf rate at time point i and depth $L = x_j\,\mathrm{LAI}$. The daily potential assimilation in sugar is $P = G \cdot \texttt{TMNFTB}(\bar{T}_{\min,7d}) \cdot 30/44$, the last factor converting CO_2 to CH_2O.

4.4 Water stress and the soil water balance

Maximum transpiration and the water-stress factor are

$$T_{\max} = ET_0\left(1 - e^{-0.75\, k\, \mathrm{LAI}}\right), \tag{7}$$

$$p = \frac{1}{0.76 + 1.5\, ET_0} - (5 - \texttt{DEPNR}) \cdot 0.1, \qquad \theta_{cr} = (1-p)(\theta_{fc} - \theta_{wp}) + \theta_{wp}, \tag{8}$$

$$f_w = \mathrm{clip}\!\left(\frac{\theta - \theta_{wp}}{\theta_{cr} - \theta_{wp}},\ 0,\ 1\right), \tag{9}$$

with θ the root-zone moisture, θ_{fc} field capacity and θ_{wp} wilting point. Actual transpiration is $T_a = f_w T_{\max}$ and actual assimilation $P \cdot f_w$. The classic water balance is a bucket over the rooted depth R (growing `RRI` cm/day up to 100 cm) plus a lower zone:

$$\frac{dW}{dt} = \text{infiltration} - T_a - E_s - \text{percolation}, \qquad \theta = W/R, \tag{10}$$

where water above field capacity percolates to the lower zone and roots gain the lower zone's water as they deepen. With no crop, R is the top 10 cm.

4.5 Respiration, growth and partitioning

$$M = \big(r_R W_R + r_L W_L + r_S W_S + r_O W_O\big)\,\texttt{RFSETB}(D)\,Q_{10}^{(T-25)/10}, \tag{11}$$

$$\Delta W_{\text{tot}} = C\,(P f_w - M), \qquad C = \left[\left(\tfrac{F_L}{c_L} + \tfrac{F_S}{c_S} + \tfrac{F_O}{c_O}\right)(1 - F_R) + \tfrac{F_R}{c_R}\right]^{-1}, \tag{12}$$

with organ weights W, maintenance coefficients r, conversion efficiencies c, and partitioning fractions F_R, F_L, F_S, F_O from tables of D. The grain grows by $\Delta W_O = F_O(1 - F_R)\,\Delta W_{\text{tot}}$, and the yield is $Y = \sum_t \Delta W_O$ (grain dry matter, `TWSO`).

4.6 Nitrogen (WOFOST 8.1, classic N balance)

Each organ demands N to reach its maximum concentration $n^{\max}$:

$$D_N = \sum_{\text{organs}} \Big[\max\big(n^{\max} W - N,\ 0\big) + n^{\max}\,\Delta W\Big]. \tag{13}$$

Legumes fix a fraction of the demand, and the rest is taken from the soil's mineral N pool N_{av}, but only if the crop is not severely water-stressed:

$$F_N = \texttt{NFIX_FR} \cdot D_N, \qquad U_N = \begin{cases} \min\big(D_N - F_N,\ N_{av},\ \texttt{RNUPTAKEMAX}\big), & f_w > 0.01, \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

Faba bean has `NFIX_FR`=0.8; maize and barley 0. The soil pool changes as

$$\frac{dN_{av}}{dt} = \eta\, N_{\text{fert}} + \text{mineralization} - U_N, \tag{15}$$

with fertilizer recovery η. Low leaf N lowers $A_{\max}$, slows leaf expansion and speeds leaf death.

5 Mexican context added to the model

5.1 Rainfed sowing

Crops are sown when the rains start: on the first day from the crop's earliest sowing date t_0 with at least 20 mm of rain over the previous five days, or after 45 days anyway:

$$t_{\text{sow}} = \min\Big\{t \ge t_0 : \sum_{k=0}^{4} P(t-k) \ge 20 \text{ mm}\Big\}, \quad t_{\text{sow}} \le t_0 + 45. \tag{16}$$

A fixed-date alternative ($t_{\text{sow}} = t_0$) is available.

5.2 Crops and fertilizer

WOFOST's spring-sown wheat is not available (its wheat parameters are winter wheats that need vernalization), and common bean (*frijol*), oats and intercropping (*milpa*) are not modeled.

Table 2: Crops and management. Fertilizer recovery $\eta = 0.5$ for all applications.

Crop	WOFOST variety	Earliest sowing	N (kg/ha)	Min. grain N	Basis
Maize	`Grain_maize_202`	May 15	80 + 80 (day 0, 40)	1.0%	160 kg N/ha: highest dose tested on rainfed hybrids at Texcoco (Alonso-Sánchez et al., 2023); split [assumption]
Barley	`Spring_barley_301`	July 1	60 (day 0)	1.2%	[assumption] no highland source found
Faba bean	`Faba_bean_801`	June 15	40 (day 0)	3.5%	[assumption]

5.3 Grain-N constraint

WOFOST 8.1 does not bound the N concentration of the grain: in a test without fertilizer, maize made 14.5 t/ha of grain holding 0.26% N, which real maize does not do (typically 1.0–1.5%). We therefore cap the N-limited yield by the N actually allocated to the grain:

$$Y_{wn} = \min\left(Y_{wn}^{\mathrm{WOFOST}},\ \frac{N_{\mathrm{grain}}}{n_{\min}}\right), \tag{17}$$

with $n_{\min}$ from Table 2. Both values are kept in the outputs.

5.4 Soil nitrogen supply

Mineralization of soil organic N is taken from IGBP-DIS total N in 0–100 cm (12,709 kg N/ha at Texcoco):

$$M_N = \mu\, N_{\mathrm{total}}, \qquad \mu = 0.005\ \mathrm{yr}^{-1} \Rightarrow M_N \approx 64\ \mathrm{kg\ N/ha\ per\ season}, \tag{18}$$

[assumption] μ, released evenly over the first 150 days of each season (March–July).

5.5 Chaining seasons: water and nitrogen carryover

Each season (March 1 to end of February) is simulated separately, and its end state starts the next one:

$$\mathrm{WAV}_{s+1} = W_{\mathrm{end},s}^{\mathrm{profile}} - R_{\max}\,\theta_{wp}, \qquad \theta_{s+1}^{\mathrm{top}} = \mathrm{clip}(\theta_{\mathrm{end},s}^{\mathrm{top}},\ \theta_{wp},\ \theta_{sat}), \tag{19}$$

$$N_{av,s+1} = \big(N_{av,\mathrm{end},s} + \rho\, N_{\mathrm{res},s}\big)(1-\lambda), \qquad N_{\mathrm{res}} = N_{RT} + (W_{RT}^{\mathrm{dead}})\, n_{RT}^{\mathrm{res}} + (W_{LV}^{\mathrm{dead}})\, n_{LV}^{\mathrm{res}}, \tag{20}$$

where the total profile water is conserved exactly across the boundary (checked numerically), N_{res} is the N in roots and dead leaves (stems and green leaves, the *rastrojo*, are assumed removed for fodder, as is common in Mexico), $\rho = 0.5$ the share of residue N released the next season [assumption] , and $\lambda = 0.5$ the share of carried mineral N lost before the next crop [assumption] . Fertilizer recovery $\eta = 0.5$ follows the 30–50% first-season recovery typical of cereals (Cassman et al., 2002). The CO_2 concentration is $380 + 2.2\,(y - 2005)$ ppm.

5.6 Frost

The crop ends on the first night, from 30 days after sowing, with

$$T_{\min}(t) + \delta \le 0\,^{\circ}\mathrm{C}, \tag{21}$$

where δ (default 0) lets us represent valley floors colder than the 0.5° POWER grid ($\delta = -2$ °C in one scenario). POWER shows on average 5.4 frost days per year at Texcoco (1990–2025), with first autumn frosts between mid-October and late December and none at all in 2021–2024.

5.7 Drought indicators and scenarios

For each season we record: rain from sowing to harvest; July–August rain (*canícula*); the longest run of days with < 1 mm of rain; water-stress days ($f_w < 0.5$); soil water on March 1. Years are classed as dry, normal or wet by terciles of May–October rain over 2005–2025. Drier climates are explored by multiplying daily rain by a factor, either all year or only in July–August.

6 Experiments

Four rotations, one crop per rainy season: maize every year; maize–barley; maize–faba bean; maize–barley–faba bean. Every phase of each rotation is simulated, so each crop occurs every year (8 sequences $\times$ 21 seasons, 2005–2025, at three production levels). The first season is a spin-up and is excluded. Scenarios: baseline; 20% and 40% less rain; 50% less rain in July–August; frost check 2 °C colder; no fertilizer; fixed sowing on each crop's earliest date.

7 Results

7.1 Baseline: where the yield goes

Table 3: Baseline results at Texcoco, 2006–2025 (20 seasons per crop and rotation). Yields are grain or seed dry matter in t/ha: Y_{wn} water- and N-limited (with the grain-N constraint), Y_w water-limited, Y_p potential. CV: coefficient of variation of Y_{wn} across years. Failed: seasons with $Y_{wn} < 1$ t/ha. Frost: seasons ended by frost before maturity. N uptake in kg N/ha.

Rotation	Crop	Y_{wn}	CV (%)	Y_w	Y_p	Water gap (%)	N gap (%)	Failed	Frost	N uptake
Maize every year	maize	6.3	42	10.8	11.7	8	42	1	12	144
Maize–barley	barley	6.1	5	11.7	11.8	1	48	0	2	98
Maize–barley	maize	6.4	33	11.6	11.7	1	45	0	12	147
Maize–faba bean	faba bean	9.7	6	9.9	10.0	1	2	0	5	88
Maize–faba bean	maize	6.9	34	11.5	11.7	2	40	0	12	154
Maize–barley–faba bean	barley	6.1	5	11.7	11.8	1	48	0	2	98
Maize–barley–faba bean	faba bean	9.8	4	10.0	10.0	0	2	0	5	87
Maize–barley–faba bean	maize	6.8	31	11.8	11.7	0	42	0	12	155

Table 3 summarizes the baseline. Fertilized rainfed maize yields 6.3–6.9 t/ha of grain dry matter (≈7.3–8.0 t/ha at 14% moisture), the same order as the 4.8–6.5 t/ha measured for rainfed hybrids at Texcoco (Alonso-Sánchez et al., 2023). Water-limited maize would yield ≈11 t/ha and potential maize 11.7 t/ha, so **nitrogen, not water, is the largest yield gap** (40–45%); the water gap (0–8%) is concentrated in a few years. The grain-N constraint is active in 69% of maize seasons (on average −2.7 t/ha relative to raw WOFOST) and in 95% of barley seasons, and never for faba bean, whose fixation keeps its seed N high. Frost ended 12 of the 20 maize seasons before maturity: with rain-onset sowing in late May or June, the long-cycle maize is still filling grain when the first frosts arrive (late October to December). Barley and faba bean tonnages are not credible (Section 9).

7.2 Rotations

Table 4: Maize yield (Y_{wn}, t/ha) by the crop grown the season before (baseline).

Previous crop	Seasons	Mean	Min	Max
barley	20	6.4	1.4	11.7
faba bean	40	6.9	1.4	12.7
maize	20	6.3	0.0	10.1

Maize after faba bean yields 6.9 t/ha against 6.3 t/ha in monoculture (+0.6 t/ha, +10%, Table 4). Paired by year, maize after faba bean is better by more than 0.5 t/ha in 11 of 20 years and worse in 4. The gain comes from nitrogen: maize after faba bean starts the season with more mineral N (11.0 vs 7.5 kg/ha on average), because faba bean draws 80% of its N from the air and leaves N-rich roots and dead leaves. In three of the four worse years (2006, 2013, 2017), maize after faba bean started with wetter soil, grew a larger canopy and spread its N over more leaves and stems, leaving less N for the grain; this response runs through the grain-N constraint and should be treated as uncertain. In 2024, maize after maize benefited from 74 kg/ha of fertilizer N left unused after the 2023 failure.

The largest single difference is 2023, the driest season in the record (Table 6): maize after maize failed (0.0 t/ha) while maize after faba bean yielded 12.7 t/ha. In 2022 the rains came late, maize was sown on 15 June and only matured on 5 January, drying the profile; on 1 March 2023 it held 146 mm against a wilting point of 137 mm. Maize sown on 15 May then met a nearly rainless June (12 mm) and spent 117 days under water stress.

7.3 Drought: amount versus timing

Table 5: Mean maize yield (Y_{wn}, t/ha) by year class. Classes are terciles of May–October rain over 2005–2025; ranges in mm.

Rotation	Dry (478–747 mm, 7 yr)	Normal (754–814 mm, 7 yr)	Wet (846–1005 mm, 7 yr)
Maize every year	5.2	5.5	8.0
Maize–barley	7.1	5.1	7.0
Maize–faba bean	7.8	5.4	7.7
Maize–barley–faba bean	7.5	5.4	7.7

Table 6: Year by year, maize every year (baseline): rain from sowing to harvest (mm), July–August rain (mm), longest run of days with < 1 mm rain during the season, days with water-stress factor < 0.5, soil water in the 0–100 cm profile on March 1 (mm; wilting point = 137 mm), and maize yield (Y_{wn}, t/ha) in monoculture vs after faba bean.

Year	Class	Sowing	Season rain	Jul–Aug rain	Dry spell	Stress days	Soil water Mar 1	Maize (mono)	Maize after faba	Frost end
2006	normal	May 19	746	313	6	15	195	10.1	8.1	yes
2007	wet	May 26	796	392	6	13	365	7.2	8.0	yes
2008	normal	May 19	747	310	17	13	355	7.2	7.3	yes
2009	normal	May 16	827	219	9	12	289	6.8	7.4	
2010	normal	Jun 12	736	453	13	13	362	1.7	1.7	yes
2011	normal	Jun 21	692	378	5	10	274	1.4	1.4	yes
2012	normal	Jun 04	744	406	26	12	343	4.7	4.8	yes
2013	wet	May 15	992	318	6	18	193	10.1	8.0	yes
2014	wet	May 15	908	269	9	12	358	6.9	7.8	yes
2015	dry	May 15	699	213	7	10	302	6.6	7.3	
2016	normal	Jun 03	779	328	17	10	319	6.5	7.1	
2017	wet	May 29	820	363	10	12	227	9.7	7.6	yes
2018	dry	Jun 10	732	263	12	10	281	6.4	7.2	yes
2019	dry	Jun 03	684	261	16	9	281	6.4	7.0	
2020	dry	Jun 11	558	280	16	13	299	5.1	5.2	yes
2021	wet	May 15	802	286	15	9	273	7.0	7.6	
2022	dry	Jun 15	535	246	22	17	252	6.8	7.5	
2023	dry	May 15	424	266	28	117	146	0.0	12.7	
2024	wet	Jun 20	864	383	16	10	200	7.9	7.0	
2025	wet	May 25	963	296	13	10	199	7.1	7.7	yes

Total season rain is a weak predictor of maize yield here (Table 5). Monoculture maize did only slightly worse in dry years (5.2 t/ha) than in normal years (5.5 t/ha), and rotations did *better* in dry years than in normal ones. Two reasons appear in Table 6: frost ended 71% of normal and wet seasons but only 33% of dry ones, and what hurts is *when* rain fails rather than how much falls. 2023 (478 mm of May–October rain, a 28-day dry spell, 12 mm in June) failed after maize;

2022 (565 mm) did not. Rotations protected maize in dry years (7.1–7.8 vs 5.2 t/ha), when the soil water and nitrogen left by the previous crop matter most.

7.4 Scenarios

Table 7: Mean maize yield (Y_{wn}, t/ha) by scenario and rotation, 2006–2025; in parentheses, the number of failed maize seasons (< 1 t/ha).

Scenario	Maize every year	Maize–barley	Maize–faba bean	Maize–barley–faba bean
Baseline	6.3 (1)	6.4 (0)	6.9 (0)	6.8 (0)
20% less rain	5.3 (1)	6.0 (1)	6.4 (0)	6.6 (0)
40% less rain	2.4 (10)	2.8 (8)	2.6 (9)	3.0 (9)
Canícula: −50% rain Jul–Aug	5.8 (2)	6.0 (1)	6.5 (1)	6.5 (0)
Frost check 2 °C colder	5.7 (1)	5.9 (1)	6.5 (1)	6.5 (1)
No fertilizer	1.9 (1)	1.9 (0)	2.4 (0)	2.3 (0)
Fixed sowing (May 15)	8.5 (0)	8.3 (0)	9.0 (0)	9.0 (0)

- **20% less rain**: maize monoculture loses 1.0 t/ha; rotations lose 0.3–0.5 t/ha. Rotations buffer moderate drying.
- **40% less rain**: 8–10 of 20 maize seasons fail in every rotation; rotation cannot compensate.
- **Stronger canícula** (half the July–August rain): −0.3 to −0.5 t/ha. At this site the soil stores 236 mm of usable water and buffers a weaker mid-summer, unlike shallow soils further north.
- **Colder nights** (frost check −2 °C): frost ends 85% of maize seasons; −0.4 to −0.6 t/ha.
- **No fertilizer**: maize falls to 1.9–2.4 t/ha, faba bean is unaffected. The Texcoco trial found only a small response (5.2 t/ha without fertilizer vs 5.6 with 160 kg N/ha; Alonso-Sánchez et al., 2023), so the model's soil N supply (μ) is too low for that site: reproducing the trial needs about 95–100 kg N/ha per season instead of 64, and about 40% less photosynthesis than the European crop parameters (see the companion calibration report, `calibration.pdf`).
- **Fixed sowing on May 15**: maize rises to 8.3–9.0 t/ha. Earlier sowing lengthens the season and escapes some frost. WOFOST does not represent failed emergence in a dry seedbed, so this gain is optimistic.

7.5 Soil moisture against SMAP

The simulated 0–100 cm soil moisture (Maize every year, 2023–2024) correlates with SMAP root-zone moisture at $r = 0.87$ over 731 days. The day-to-day dynamics agree, but levels differ: in the 2023 rainy season the model's profile (0.13–0.19) was drier than SMAP's (0.27–0.32), as expected for a field whose previous crop dried it, compared with a 9 km cell with its own soil description.

8 Verification

The core daily equations shared by WOFOST 7.2 and 8.1 (phenology, light interception and Gaussian integration, water stress, maintenance respiration, conversion and partitioning, Penman–Monteith ET_0) were recomputed by hand for one simulated day (7 August 2023, maize after barley at Texcoco, WOFOST 7.2) and matched PCSE to all printed decimals: $ET_0 = 5.108$ mm, $dD/dt = 0.013442$, $G = 943.79$ kg CO_2/ha, $f_w = 0.916$, $T_a = 3.821$ mm, $M = 138.82$ kg CH_2O/ha, $\Delta W_O = 302.36$ kg/ha. The season's grain yield equals the sum of daily grain growth (17,700 kg/ha). The nitrogen-specific equations of WOFOST 8.1 have not yet been verified by hand.

9 Limitations

- **Crop parameters are European.** Absolute yields of barley and especially faba bean are not credible (simulated faba bean seed ≈10 t/ha vs. 1–2 t/ha typical in Mexico); compare maize across rotations, not the partner crops' tonnages. Maize yields are in the range of a Texcoco trial (Alonso-Sánchez et al., 2023) but the model is not calibrated.
- **Several nitrogen parameters are assumptions** ($\mu, \rho, \lambda, \eta, n_{\min}$, barley and faba bean doses); results on rotation benefits depend on them.
- **No pests, diseases, weeds, phosphorus, hail or lodging**; no tillage or residue-cover effects on evaporation.
- **Weather is coarse.** The POWER cell's elevation is 2,633 m vs. ≈2,250 m at the site; valley-floor frosts are under-represented.
- **Frost ends the crop on a single threshold**; WOFOST has no partial frost damage for these crops.
- **Soil hydraulic conductivity and percolation** use PCSE defaults, not site data.
- **One site.** Results do not transfer to other soils or climates without rerunning.
- The rain-scaling scenarios change only rain amounts, not temperature or radiation.

10 Next steps

1. Calibrate maize (and partner crops) against observed yields: INIFAP/CIMMYT trials at Texcoco, SIAP municipal statistics.
2. Replace or bias-correct POWER temperatures with station data (SMN, Chapingo) for frost.
3. Use SMAP root-zone moisture to initialize or correct soil water at sowing.
4. Add supplemental irrigation as a drought adaptation option.
5. Add prices and costs to score rotations, then frame rotation choice as a sequential decision problem for reinforcement learning (CropGym; Kallenberg et al., 2023; cf. CyclesGym, Turchetta et al., 2022).

11 Reproducibility

Code: `src/data_ingestion/` and `src/crop_models/wofost_rotations.py`. Python 3.13, PCSE 6.0.13. From the repository root:

```
python -m src.data_ingestion.smap --lat 19.53 --lon -98.85 --start 2023-01-01 \
    --end 2024-12-31 --variables sm_rootzone sm_surface sm_rootzone_wetness \
    --out data/smap_texcoco_2023_2024.csv
M=src.crop_models.wofost_rotations; L=data/rotations
python -m $M --smap data/smap_texcoco_2023_2024.csv --out $L/baseline.csv
python -m $M --rain-factor 0.8 --out $L/rain80.csv
python -m $M --rain-factor 0.6 --out $L/rain60.csv
python -m $M --rain-factor 0.5 --rain-months 7 8 --out $L/canicula50.csv
python -m $M --frost-tmin-offset -2 --out $L/frost_minus2.csv
python -m $M --n-scale 0 --out $L/no_fertilizer.csv
python -m $M --sowing fixed --out $L/fixed_sowing.csv
python docs/technical_report/make_tables.py
```

References

Richard G. Allen, Luis S. Pereira, Dirk Raes, and Martin Smith. Crop evapotranspiration: Guidelines for computing crop water requirements. FAO Irrigation and Drainage Paper 56, Food and Agriculture Organization of the United Nations, Rome, 1998.

H. Alonso-Sánchez, M. Tadeo-Robledo, A. Espinosa-Calderón, J. Zaragoza-Esparza, C. López-López, B. Zamudio-González, et al. Efecto de la densidad de población y la fertilización sobre la productividad del agua y rendimientos de híbridos de maíz en el Valle de México. *Terra Latinoamericana*, 41:e1577, 2023. doi: 10.28940/terra.v41i0.1577.

Kenneth G. Cassman, Achim Dobermann, and Daniel T. Walters. Agroecosystems, nitrogen-use efficiency, and nitrogen management. *Ambio*, 31(2):132–140, 2002. doi: 10.1579/0044-7447-31.2.132.

Allard de Wit. PCSE: Python crop simulation environment, version 6.0. https://pcse.readthedocs.io, 2024a.

Allard de Wit. WOFOST crop parameters repository. https://github.com/ajwdewit/WOFOST_crop_parameters, 2024b.

Allard de Wit, Hendrik Boogaard, Davide Fumagalli, Sander Janssen, Rob Knapen, Daniel van Kraalingen, Iwan Supit, Raymond van der Wijngaart, and Kees van Diepen. 25 years of the WOFOST cropping systems model. *Agricultural Systems*, 168:154–167, 2019. doi: 10.1016/j.agsy.2018.06.018.

Global Soil Data Task. Global gridded surfaces of selected soil characteristics (IGBP-DIS). ORNL DAAC, Oak Ridge, Tennessee, USA, 2000.

Michiel G. J. Kallenberg, Hiske Overweg, Ron van Bree, and Ioannis N. Athanasiadis. Nitrogen management with reinforcement learning and crop growth models. *Environmental Data Science*, 2, 2023. URL https://www.cambridge.org/core/product/358749FAFAA4990B1448DAB7F48D641C.

NASA Langley Research Center. POWER project: Prediction of worldwide energy resources, daily data, agroclimatology community. https://power.larc.nasa.gov, 2026. Accessed October 2026.

National Snow and Ice Data Center. User notice: SMAP data reprocessing. https://nsidc.org/data/user-resources/data-announcements/user-notice-smap-data-reprocessing, 2026. Accessed October 2026.

R. Reichle, G. De Lannoy, R. D. Koster, W. T. Crow, J. S. Kimball, Q. Liu, and M. Bechtold. SMAP L4 global 3-hourly 9 km EASE-Grid surface and root zone soil moisture geophysical data, version 8 (SPL4SMGP). NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado, USA, 2025. Accessed October 2026.

Rolf H. Reichle, Gabrielle J. M. De Lannoy, Qing Liu, Joseph V. Ardizzone, Andreas Colliander, et al. Assessment of the SMAP level-4 surface and root-zone soil moisture product using in situ measurements. *Journal of Hydrometeorology*, 18(10):2621–2645, 2017. doi: 10.1175/JHM-D-17-0063.1.

M. Rodell, P. R. Houser, U. Jambor, J. Gottschalck, K. Mitchell, et al. The global land data assimilation system. *Bulletin of the American Meteorological Society*, 85(3):381–394, 2004. doi: 10.1175/BAMS-85-3-381.

C. J. T. Spitters, H. A. J. M. Toussaint, and J. Goudriaan. Separating the diffuse and direct component of global radiation and its implications for modeling canopy photosynthesis. Part I. components of incoming radiation. *Agricultural and Forest Meteorology*, 38:217–229, 1986. doi: 10.1016/0168-1923(86)90060-2.

Iwan Supit, A. A. Hooijer, and C. A. van Diepen. *System description of the WOFOST 6.0 crop simulation model implemented in CGMS*. Office for Official Publications of the European Communities, Luxembourg, 1994.

Matteo Turchetta, Luca Corinzia, Scott Sussex, Amanda Burton, Juan Herrera, Ioannis N. Athanasiadis, Joachim M. Buhmann, and Andreas Krause. Learning long-term crop management strategies with CyclesGym. In *Advances in Neural Information Processing Systems 35, Datasets and Benchmarks Track*, 2022.